RESUMELENS
FORMAL LANGUAGES

screening résumés with the theory of computation

OUR PIPELINE

1. **Extract** · Regular expressions
2. **Normalize** · Finite-state transducers
3. **Classify** · Finite automata (DFA)
4. **Specify** · Context-free grammar

→ Validated HTML profile

Can formal-language theory screen résumés reliably and explainably?

ResumeLens reads a plain-text résumé and decides which job profiles the candidate fits, with no machine learning. Regular expressions extract the data, transducers normalize it, one automaton per profile accepts or rejects it, and a grammar produces a validated document. Every decision is traceable state by state.

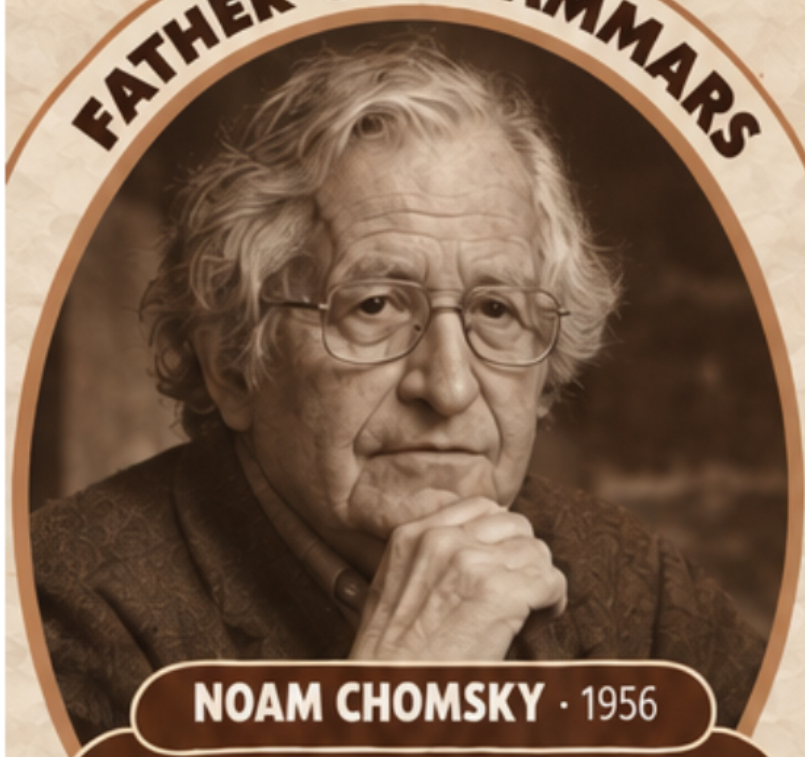


KEN THOMPSON

Turned regular expressions into working automata (1968) — our Stage 1 & 3 engine idea

883	5	4	16
tests	theorems	DFAs	grammar nonterminals

FATHER OF GRAMMARS



NOAM CHOMSKY · 1956

His hierarchy regular ⊂ context-free is the ladder of our project: Stages 1–3 live in the regular world, Stage 4 uses a context-free grammar.

$$R_{FS} = E^*\ F^*\ G^*\ (B{+}K^* \mid K^*)\ A^*\ (S{+}N^* \mid N^*)\ D^*V^*$$

E experience · **F** frontend language · **G** frontend framework · **B** backend language · **K** backend framework · **A** API · **S** SQL database · **N** NoSQL database · **D** DevOps/cloud · **V** version control



Kleene

STAGE 1 · EXTRACT — Regular expressions

Kleene, 1956: regular expressions = finite automata

- Plain-text résumé (up to 100,000 characters).
- 19 regex specs: email, phone, LinkedIn, GitHub, name, years of experience, job title, degree, certification, languages, frameworks, databases, APIs, version control, DevOps tools.
- 3 years → JS, React.js, NodeJS, Postgres, Git → years = 3 + 5 raw tokens.
- 209 tests.



Mealy

STAGE 2 · NORMALIZE — Finite-state transducers

Mealy, 1955: machines that write output on every transition

- Cascade CLEAN → ALIAS → EXPERIENCE → FILTER.
- JS → JAVASCRIPT · React.js → REACT · NodeJS → NODE_JS · Postgres → POSTGRESQL.
- 79 canonical tokens; ALIAS has 503 states and 616 transitions.
- 426 tests · 40 of 43 mutants killed.



Rabin & Scott

STAGE 3 · CLASSIFY — Deterministic finite automata

Rabin & Scott, 1959: finite automata theory

- 4 DFAs: Full Stack, Machine Learning, Cloud & DevOps, Data Scientist.
- Skills by category in canonical order; δ total, trap state q_trap.
- 5 proved theorems, minimality via Moore's algorithm.
- 118 tests · 100 % coverage · 35 of 37 mutants killed.

WEDNESDAY ADDAMS



Backus

STAGE 4 · SPECIFY — Context-free grammar

Backus & Naur, 1960: BNF notation

- Our own DSL. 16 nonterminals, 6 fixed sections (personal, contact, experience, education, skills, classification).
- Stages 1–3 generate the DSL; text validates syntax and rules.
- Rendered as a safe HTML page.
- 87 tests.

1 RAW TEXT

Wednesday Addams
3 years of experience developing web applications.
Technical Skills: JS, React.js, NodeJS, Postgres, Git.

2 TRANSDUCERS

EXPERIENCE_PRESENT · JAVASCRIPT · REACT · NODE_JS · POSTGRESQL · GIT

3 DFAs

Full Stack ✓ accepted
Machine Learning ✗
Cloud & DevOps ✗
Data Scientist ✗

4 DSL

```
candidate {
  personal { name "Wednesday Addams"; }
  experience { total_years 3; }
  skills { skill JAVASCRIPT; skill REACT;
      skill NODE_JS; skill POSTGRESQL;
      skill GIT; }
  classification { accepted full_stack; }
}
```

RESUMELENS VALIDATED DSL

Wednesday Addams
Experience: 3 years
Canonical skills: JAVASCRIPT, REACT, NODE_JS, POSTGRESQL, GIT
Declared accepted profiles: Full Stack Developer.

RESULTS

883	5	100%	75/80
tests all passing	theorems proved	Stage 3 coverage	mutants killed

Tests per stage:

Stage 1	209
Stage 2	426
Stage 3	118
Stage 4	87
Integration	43

Two interfaces: a Streamlit app with live DFA traces and a React web client (raw text / DSL / HTML).

Each formal tool fits one job, and our grade came from the wrong claim of ours: the Full Stack DFA was not minimal.

CONCLUSIONS & LIMITATIONS

- Rule-based: only skills in our alias dictionary are recognized.
- Next: more profiles and aliases, PDF/DOCX input.

STANDING ON THE SHOULDERS OF...

Stephen Kleene	George Mealy	Rabin & Scott	Noam Chomsky
1956	1955	1959	1960
regular expressions → Stage 1	machines with output → Stage 2	finite automata theory → Stage 3	grammars → Stage 4

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Kleene 1956 · Mealy 1955 · Moore 1956 · Chomsky 1956 · Rabin & Scott 1959 · Backus et al. 1960 · Thompson 1968 · Hopcroft, Motwani & Ullman 2006 · Photos: Wikimedia Commons · github.com/JuanJoe-18/DiscreteIII-Integrative-Task-VictorTeam